

GRAAFF-REINET, South Africa

WMO: 687370

Lat: 32.193S

Lon: 24.543E

Elev: 792

StdP: 92.17

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	-1.8	-0.3	-14.0	1.2	19.2	-11.0	1.6	16.3	9.4	17.4	8.5	18.1	1.3	10	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.7	36.0	17.7	34.1	17.4	32.4	17.3	21.4	27.9	20.5	27.1	19.8	26.6	4.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	15.8	24.2	18.6	14.9	23.1	17.7	14.0	22.6	66.4	27.8	62.8	27.2	60.2	26.5	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.5	6.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	23.8	23.6	21.3	17.8	15.0	11.8	12.8	12.7	15.7	18.4	20.3	22.4
	DBStd	5.73	3.30	2.98	3.32	4.14	4.43	4.08	4.85	4.64	3.99	4.20	3.58	3.13
	HDD10.0	82	0	0	0	1	7	26	26	20	2	1	0	0
	HDD18.3	937	3	1	10	60	125	198	182	186	99	50	18	4
	CDD10.0	2981	427	382	352	234	161	80	112	104	173	262	310	383
	CDD18.3	794	172	150	103	43	21	2	9	12	20	53	78	129
	CDH23.3	7716	1721	1303	876	343	169	11	52	125	293	647	904	1272
	CDH26.7	3180	813	558	339	106	37	0	2	22	101	270	382	550
(14) Wind	WSAvg	3.0	3.2	2.9	2.8	2.4	2.6	3.0	3.1	2.7	2.8	3.2	3.3	3.4
(15) Precipitation	PrecAvg	306	31	39	48	28	19	13	15	19	9	26	28	32
	PrecMax	580	137	128	181	98	150	52	92	100	56	100	101	124
	PrecMin	146	0	0	0	2	0	0	0	0	0	0	2	0
	PrecStd	104	32	34	42	23	28	12	22	25	12	27	24	33
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.6	37.4	35.8	32.2	29.8	24.7	26.7	29.3	32.7	36.3	36.6	37.4
		MCWB	17.9	18.5	18.1	16.0	14.3	11.4	11.1	12.7	13.7	16.2	16.2	17.0
	2%	DB	35.6	34.4	32.8	29.6	27.3	23.2	24.8	26.6	29.8	32.8	33.3	34.4
		MCWB	18.2	18.9	18.1	16.0	13.8	10.7	10.9	10.9	13.4	15.6	16.1	17.0
	5%	DB	33.4	32.1	30.4	27.1	25.0	21.5	22.9	24.4	26.9	30.0	30.9	32.1
		MCWB	18.3	18.9	17.7	15.6	12.9	10.2	10.5	10.5	12.7	14.7	16.0	17.3
	10%	DB	31.2	30.1	28.2	24.7	22.7	19.4	21.0	22.0	24.1	27.2	28.7	29.9
		MCWB	18.1	18.7	17.3	15.0	12.1	9.5	9.7	9.8	12.1	14.5	15.6	17.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.2	23.1	22.1	18.6	15.8	12.6	13.0	15.3	16.5	18.6	20.1	21.7
		MCDB	30.1	29.5	28.2	24.6	25.4	21.6	22.5	22.9	26.5	29.3	28.3	27.4
	2%	WB	20.8	21.6	20.6	17.7	14.7	11.7	11.9	13.2	15.1	17.1	18.5	20.3
		MCDB	28.7	27.6	26.3	24.7	24.2	20.3	21.1	19.4	24.9	28.0	27.7	27.0
	5%	WB	19.9	20.6	19.6	16.8	13.8	10.9	11.2	12.0	14.1	16.3	17.5	19.5
		MCDB	27.1	26.9	25.7	23.7	22.5	19.1	20.5	19.2	23.5	25.8	26.7	26.3
	10%	WB	19.2	19.8	18.6	16.0	12.9	10.1	10.3	10.8	13.0	15.4	16.7	18.6
		MCDB	27.4	26.4	25.1	22.7	20.7	17.9	19.4	19.6	22.1	24.1	25.6	25.9
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	12.5	12.5	12.4	13.3	12.7	13.6	14.4	14.1	14.2	14.3	13.9
		MCDBR	17.7	16.2	16.4	15.3	15.4	14.4	14.3	16.3	19.3	19.0	19.0	17.9
		MCWBR	4.7	4.6	5.2	5.5	6.3	6.6	6.4	6.8	7.4	6.6	5.8	5.2
	5% WB	MCDBR	13.8	12.1	12.4	12.8	14.1	12.6	13.7	13.3	15.6	15.6	15.4	13.3
		MCWBR	4.7	4.7	5.1	5.1	6.8	6.8	6.9	6.5	7.0	5.9	5.6	4.8
(40) Clear-Sky Solar Irradiance	taub		0.346	0.353	0.338	0.309	0.288	0.274	0.274	0.318	0.342	0.358	0.334	0.340
	taud		2.496	2.476	2.496	2.561	2.598	2.605	2.593	2.445	2.361	2.361	2.477	2.499
	Ebn at Noon		993	966	946	924	897	888	904	898	925	949	998	1002
	Edh at Noon		115	113	104	87	74	69	73	96	117	126	116	115
(44) All-Sky Solar Radiation	RadAvg		7.66	6.93	5.81	4.47	3.51	3.10	3.37	4.23	5.54	6.66	7.58	8.13
	RadStd		0.37	0.35	0.34	0.18	0.13	0.17	0.18	0.24	0.33	0.30	0.38	0.46

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page